

Dr. Divyesh Mistry

Computational Materials Scientist

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PROFESSIONAL SUMMARY

Computational Materials Scientist and postdoctoral researcher with 7+ years of research experience developing atomistic and multiscale models of deformation, defect evolution, and mechanical behavior in advanced materials. Specializes in large-scale molecular dynamics simulations using **LAMMPS** and high-performance computing, complemented by discrete dislocation dynamics and continuum modeling. Develops automated **Python** workflows for simulation management, trajectory processing, defect characterization, and large-scale data analysis. Research spans tungsten alloys, nickel-based superalloys, and polymeric adhesive systems, with relevance to structural materials operating under extreme thermomechanical conditions.

CORE COMPETENCIES

Atomistic & Multiscale Modelling	⚙️	Molecular Dynamics (MD), Discrete Dislocation Dynamics (DDD), Atomistic Simulations, Microstructure Evolution, Defect Mechanics
Computational Modeling Tools	🔧	LAMMPS, OVITO, AtomsK, ABAQUS, ANSYS, COMSOL
Programming & Scientific Computing	📄	Python, High-Performance Computing (HPC, MPI/OpenMP), C++, Fortran, Bash, Parallel Simulation Frameworks, Git
Data Analysis & Visualization	📊	NumPy, Pandas, SciPy, Matplotlib, Automated Post-Processing, High-Volume Simulation Data Analysis
Tools & Environments	💻	Linux (Ubuntu), High-Performance Computing Clusters, Git/GitHub, LaTeX

EXPERIENCE

2026 – Present	Postdoctoral Researcher , <i>University of Sevilla, Sevilla, Spain</i> ➤ Investigating fundamental multiscale mechanisms of damage and degradation in metal/polymer (steel/adhesive) interfaces exposed to moisture ingress and harsh marine environments. ➤ Developing computational models spanning from atomistic simulations of interfaces to coupled diffusion/fatigue models for predicting joint lifetime. ➤ Contributing to the qualification of lightweight hybrid structures and developing life assessment tables for design, maintenance, and repair protocols. ➤ Project : <i>Multiscale modeling of interfacial delamination of steel/adhesive interfaces by aqueous corrosion</i>, funded by the U.S. Office of Naval Research. ➤ Advisors : Prof. María del Pilar Ariza Moreno and Prof. Michael Ortiz.
April 2026 – July 2026	Visiting Researcher , <i>Merrimack College, Department of Mechanical and Electrical Engineering, North Andover, MA, USA</i> ➤ Conducted large-scale molecular dynamics simulations of temperature-dependent crack-tip deformation and ductile-to-brittle transition behavior in tungsten and tungsten-rhenium alloys. ➤ Developed automated LAMMPS workflows for high-performance computing clusters to examine the effects of temperature, alloy composition, and defect structure on mechanical response. ➤ Built custom Python pipelines using NumPy and Pandas for trajectory processing, stress-strain analysis, defect characterization, and visualization of large atomistic datasets. ➤ Investigated the roles of dislocation activity, crack-tip plasticity, and local structural evolution in controlling stress relaxation and fracture behavior. ➤ Complemented the atomistic analysis with a representative three-dimensional polycrystalline DDD framework for studying Frank–Read source activation, dislocation multiplication, and density-dependent hardening.

- Jan 2025 – Dec 2025** **Junior Research Fellow (DRDO Project), IIT Bombay, Aerospace Engineering, Mumbai, India**
- Conducted research on the constitutive modelling and high strain-rate response of Shape Memory Alloy (SMA)-based composites for aerospace and defense applications.
 - Performed dynamic impact experiments using Split Hopkinson Pressure Bar (SHPB) systems to characterize material behaviour under high strain-rate loading.
 - Integrated experimental observations with numerical simulations to calibrate and validate constitutive models for aerospace structural applications.
 - Assisted in project execution, data analysis, and mentoring of junior researchers in experimental characterization and computational modelling.
- Jan 2019 – June 2025** **Ph.D. Research Scholar, IIT Bombay, Aerospace Engineering, Mumbai, India**
- Designed and executed large-scale atomistic simulations using **LAMMPS** on high-performance computing platforms to investigate generalized stacking fault energies, defect mobilities, and dislocation core structures.
 - Developed a physics-informed hierarchical multiscale framework that maps atomistic-derived parameters (drag coefficients, obstacle strengths, and lattice strain fields) directly into meso-scale continuum models.
 - Engineered automated post-processing and data-mining pipelines using **Python** to parse high-volume molecular dynamics (MD) trajectory data, track defect coordinates, and analyze structural evolution via OVITO.
 - Authored a custom parallel polycrystalline Discrete Dislocation Dynamics (DDD) code coupled with Finite Element Analysis (FEA) to simulate 3D defect interactions and slip localization near internal interfaces.
 - Utilized parallel computing (MPI) to scale atomistic simulations up to multi-million atom systems, establishing quantitative relationships between alloy microstructures and mechanical response under high stress fields.
- Jan 2019 – Dec 2023** **Graduate Teaching Assistant, IIT Bombay, Aerospace Engineering, Mumbai, India**
- Assisted instruction in Continuum Mechanics, Finite Element Methods (FEM), and Multiscale Modelling for undergraduate and postgraduate engineering courses.
 - Developed and supervised computational laboratory exercises involving numerical methods, finite element analysis, and scientific programming.
- June 2016 – Dec 2018** **Assistant Professor, CMR Institute of Technology, Mechanical Engineering, Bangalore, India**
- Taught undergraduate courses in Finite Element Methods, Experimental Stress Analysis, and Solid Mechanics.
 - Developed computational laboratory modules using ANSYS, MATLAB, and Python to support engineering analysis and numerical modelling.
- June 2014 – July 2016** **Graduate Teaching Assistant & M.Tech. Researcher, IIT Kharagpur, Mechanical Engineering, Kharagpur, India**
- Developed and validated computational models using ABAQUS and ANSYS for structural mechanics and vibration analyses.
 - Conducted laboratory instruction in Finite Element Modelling, Vibration Analysis, and Tribology, integrating theoretical concepts with numerical simulations.

EDUCATION

- Jan 2019 – Dec 2025**  **INDIAN INSTITUTE OF TECHNOLOGY - BOMBAY, MUMBAI, INDIA**
- Doctorate, Aerospace Engineering
Thesis : Multiscale Modeling of Prior Particle Boundaries in Nickel-based Superalloys
CGPA : 9.36/10 *Funding* : MHRD Teaching Assistantship through Project
Key Subjects : Multiscale Modeling, Continuum Mechanics, Plasticity, Computational Mechanics
Advisors : Prof. Amuthan A. Ramabathiran, Prof. P. J. Guruprasad

- Jun 2014 – Jul 2016**  INDIAN INSTITUTE OF TECHNOLOGY - KHARAGPUR, WEST BENGAL, INDIA
M.Tech., Mechanical Engineering (Mechanical System Design)
Thesis : A Mesh-Free Model for Static Analysis of Smart Composite Beams
CGPA : 8.85/10 *Funding* : MHRD GATE Fellowship
Key Subjects : Solid Mechanics, Fracture Mechanics, Finite Element Methods, Composites, Elasticity, Vibration
Advisor : Prof. Manas Chandra Ray
- Dec 2013**  THE AERONAUTICAL SOCIETY OF INDIA, NEW DELHI
B.E., Aeronautical Engineering (Aero Mechanical)
Key Subjects : Aircraft Structures, Aerodynamics, Propulsion, Stability & Control, Solid and Fluid Mechanics

PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

- [1] **Mistry, D. A.** & Ramabathiran, A. A.,
Size-Dependent Power Laws for Edge Dislocations in Nickel Superalloys : A Molecular Dynamics Study,
Computational Materials Science, 259, 114122 (2025). DOI : 10.1016/j.commatsci.2025.114122.
- [2] **Mistry, D. A.**, Tak, T. N., Sankarasubramanian, R., & Guruprasad, P. J.,
A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-based Superalloys,
Journal of Engineering Materials and Technology, pp. 1–55 (2026). Published online : August 4, 2026. DOI : 10.1115/1.4072448.
- [3] **Mistry, D. A.** & Mahata, A.,
Crack-Tip Plasticity and Ductile-to-Brittle Transition in W and W-Re Systems through Atomistic Fracture and Mesoscale Dislocation Dynamics.
Manuscript in Preparation, 2026.
- [4] **Mistry, D. A.**, Tak, T. N., & Guruprasad, P. J.,
Polycrystalline Discrete Dislocation Dynamics of Ni-Based Superalloys : Interactions with Prior Particle Boundaries and Second Phases Using Atomistically Informed Inputs.
Manuscript in Preparation.
- [5] **Mistry, D. A.**, Mahata, A., Ramabathiran, A. A., Guruprasad, P. J., & Tak, T. N.,
Multiscale Modeling of Dislocation–Precipitate–PPB Interactions in Powder-Metallurgy Nickel Superalloys.
Engineering Mechanics Institute Conference (EMI 2026), June 2–5, 2026, Boulder, Colorado, USA.
- [6] **Mistry, D. A.**, Tak, T. N., & Guruprasad, P. J.,
Hierarchical Multiscale Modeling of Plasticity in Ni-Based Superalloys : The Combined Role of Prior Particle Boundaries and Second Phases.
14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), October 12–16, 2025, IIT Roorkee, India.
- [7] **Mistry, D. A.**, Guruprasad, P. J., & Ramabathiran, A. A.,
An Atomistically Informed Discrete Dislocation Dynamics Study of Prior Particle Boundaries in Ni-Based Superalloys.
17th Conference on Computational Plasticity (COMPLAS), September 5–7, 2023, Barcelona, Spain.

CERTIFICATIONS

- Jan 2020**  **Using Python for Research**, edX, HarvardX, Harvard University.
- Aug 2020**  **The Unix Workbench**, Coursera, Johns Hopkins University.
- Aug 2020**  **Material Behavior**, Coursera, Georgia Institute of Technology.
- May 2018**  **A Hands-on Introduction to Engineering Simulations**, edX, CornellX, Cornell University.

AWARDS, ACHIEVEMENTS, AND HONORABLE MENTIONS

- 2025**  **Membership** : Member of The Minerals, Metals & Materials Society (TMS), USA
- 2025**  Institution of Eminence Funds, IIT Bombay : Awarded to attend the international conference *TMS 2025*, USA
- 2023**  IIT Bombay Travel Grant : Awarded to attend the international conference *COMPLAS 2023*, Spain

- 2019**  MHRD Teaching Assistantship Through Project : Awarded by the **Industrial Research and Consultancy Centre (IRCC)**
- 2014**  **All India Rank 69** : Achieved in the Graduate Aptitude Test in Engineering (GATE)
- 2013**  **Permanent Membership** : Awarded permanent membership as an **Associate Member** at the **Aeronautical Society of India**